Efficient Attention Transformer Network With Self-Similarity Feature Enhancement for Hyperspectral Image Classification

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Abstract—**Recently, transformer has gained widespread application in hyperspectral image classification (HSIC) tasks due to its powerful global modeling ability. However, the inherent high-dimensional property of hyperspectral images (HSIs) leads to a sharp increase in the number of parameters and expensive computational costs. Moreover, self-attention operations in transformer-based HSIC methods may introduce irrelevant spectral–spatial information, and thus may consequently impact the classification performance. To mitigate these issues, in this article, we introduce an efficient deep network, named efficient attention transformer network (EATN), for practice HSIC tasks. Specifically, we propose two self-similarity descriptors based on the original HSI patch to enhance spatial feature representations. The center self-similarity descriptor emphasizes pixels similar to the central pixel. In contrast, the neighborhood self-similarity descriptor explores the similarity relationship between each pixel and its neighboring pixels within the patch. Then, we embed these two self-similarity descriptors into the original patch for subsequent feature extraction and classification. Furthermore, we design two efficient feature extraction modules based on the preprocessed patches, called spectral interactive transformer module and spatial conv-attention module, to reduce the computational costs of the classification framework. Extensive experiments on four benchmark datasets show that our proposed EATN method outperforms other state-of-the-art HSI classification approaches.**

***Index Terms*—Attention, hyperspectral image classification (HSIC), self-similarity, spectral interactive, transformer.**

I. Introduction

HYPERSPECTRAL images (HSIs) contain abundant spectral information from hundreds of continuous bands, including visible and infrared light [1]. Thus, they can provide comprehensive information on ground objects by accurately reflecting their spectral signatures and spectral reflectances. Nowadays, HSIs have been successfully applied in various practical fields, such as geological exploration [2], environmental monitoring [3], and agriculture [4]. Furthermore, hyperspectral image classification (HSIC) serves as the foundation of numerous geophysical exploration tasks, aiming to assign an appropriate label to each pixel in HSIs.

In the early stages, various traditional methods, such as support vector machines (SVM) [5], k-nearest neighbor [6], sparse representation [7], random forests [8], morphological profiles [9], have achieved considerable performance in HSIC tasks. However, since the same material may exhibit different spectral signatures under different circumstances, relying solely on statistical analysis and prior information obtained from training data can be susceptible to noises. Therefore, they fail to achieve promising classification performance in most cases. To mitigate this issue, many studies [10], [11] attempt to improve the performance of the HSIC models by simultaneously considering the spatial and spectral information of individual pixels.

Deep learning has been widely applied in HSIC tasks in the past decade. Specifically, convolutional neural networks (CNNs) have been employed to extract local spatial and spectral information from HSIs by utilizing a series of convolutional kernels with different sizes. By leveraging their ability to extract local information from HSIs, these networks significantly enhance the classification performance of HSIs. Hu et al. [12] introduced CNNs into the HSIC task. However, this network based on 1-D-CNN only extracted HSI features in the spectral domain, failing to utilize the spatial information of HSIs. To solve this issue, several works [14], [15] have investigated 2-D CNNs in HSIC applications. Song et al. [14] proposed a deep feature fusion network by integrating residual connections into the CNN-based HSIC method, which improves classification performance by effectively fusing multilayer feature information. In [15], a 2-D CNN-based HSIC model was proposed by extracting spatial–spectral features of HSIs. Due to the 3-D property of HSIs, Li et al. [16] designed a 3-D CNN model to extract spatial–spectral information from HSIs without preprocessing or postprocessing. However, the accumulation of 3-D CNNs generates a large number of parameters, resulting in expensive computational costs. To mitigate this issue, Roy et al. [11]

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The source code of this work is available at https://github.com/szq0816/EATN.

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presented a hybrid spectral CNN by combining 2-D CNNs and 3-D CNNs, significantly reducing the complexity of pure 3-D-based models. Although CNNs leverage their built-in inductive biases to achieve strong classification performance, they often struggle to establish global dependencies. This limitation arises from the restricted receptive fields of individual convolutional layers.

To address these limitations of CNNs, transformer has emerged as a promising solution for modeling global dependencies. Recently, transformer has been effectively adapted to various practice tasks, resulting in significant performance improvements. [17], [18], [19] A key component of transformer is self-attention, which assigns a weight to each position in a sequence by computing a weighted average of all feature vectors. This mechanism enables the model to effectively extract long-range information from images and adaptively aggregate spatial information. Vision transformer (ViT) [20] represents the pioneering attempt to employ a pure transformer model for visual tasks. It divides the input image into several patches, then projects each patch into a fixed-length vector, and feeds it to the transformer. For the convenience of classification tasks, a special token is added to the input sequence, and the corresponding output for this token serves as the final category prediction. Hong et al. [21] introduced SpectralFormer by applying the pure transformer structure to HSIC tasks. Peng et al. [22] proposed a dual-branch transformer method that uses cross-attention to facilitate interaction between tokens from the spectral and spatial branches. These methods follow the token-based ViT classification approach in practice problems. Recently, several works [23], [24] have utilized only the encoder part of the transformer networks in real applications. Mei et al. [24] argued that representing individual pixels as tokens is more suitable for patch-based HSIC tasks, and thus introduced pixel group embedding in the proposed transformer-based network. Considering the advantages of the CNNs and transformer models, several hybrid networks have emerged over the past few years. Zhao et al. [25] combined center position encoding with the CNN and transformer block, allowing the network to model local–global dependences while effectively utilizing the position information of the pixels and the spectral information. Xu et al. [18] proposed a dual-branch convolution-transformer network, incorporating a ConvTE block to effectively leverage the strengths of CNN and transformer.

Although the aforementioned transformer-based networks have shown impressive performance in HSIC tasks, they still encounter the following limitations. 1) In transformer-based networks, the elementwise calculations in multihead self-attention lead to a significant number of parameters, rendering them unsuitable for real-time applications. 2) The self-attention mechanism in transformer focuses on capturing global information, so it may overlook the importance of central pixel and positional information, which are crucial for classification.

Several studies have demonstrated that adjacent pixels in HSI patches possess different discriminative capabilities. This implies that pixels at different positions may hold distinct importance for classification tasks. To address this issue, the attention mechanism is introduced into HSCI tasks. Zhu et al. [26] designed separate spectral and spatial attention modules and subsequently added them to the CNN-based classification network. Shu et al. [27] designed split attention to fuse multireceptive field information. It effectively suppresses useless information while emphasizing useful pixels and bands. However, these HSIC methods only classify the central pixel of the patch, and its adjacent pixels play an auxiliary role in the classification. To solve this issue, Xue et al. [28] designed a local transformer module to explore the relationship between the central pixel and neighboring pixels. Similarly, Zhang et al. [29] emphasized useful spatial features by computing the cosine similarity between central and adjacent vectors. However, the positional information in the original input patch may be lost during network learning. Therefore, it is particularly critical to explore the pixel relationship in the original HSI patch. Although some existing works [30] seek to explore the relationship between the central spectral vector and adjacent spectral vectors of the original patch, they ignore local positional relationships between adjacent pixels.

To address the aforementioned issues, we propose an efficient attention transformer network (EATN) for the HSIC problem in this article. For the original HSI patches, we design a self-similarity feature enhancement (SSFE) module to generate two descriptors, namely the center self-similarity descriptor and the neighborhood self-similarity descriptor. These two descriptors are fused with the original patch to enhance the representation of spatial features from different perspectives. This strategy renders the original patch more suitable for feature extraction. Moreover, we introduce two efficient feature extraction modules, called the spectral interactive transformer (SIT) module and the spatial conv-attention (SCA) module. Specifically, the SIT module employs a group interactive spectral self-attention (GISSA) module to extract spectral information, effectively mitigating the computational burden when applying self-attention to all bands. In the SCA module, we alternately use convolution and self-attention modules to extract local and global spatial features of HSIs. Moreover, we share attention maps between SCA modules, effectively addressing the redundancy issue in self-attention computation. Extensive experiments on several benchmark HSI datasets have demonstrated the effectiveness and efficiency of the proposed EATN method in HSIC tasks.

The main contributions of this article are summarized as follows.

1) We propose a novel deep framework, called EATN, for the HSIC problem. By leveraging a feature enhancement strategy, the proposed model attempts to combine CNNs and transformer to efficiently and accurately classify each pixel of HSIs.
2) For the model's input, we design the SSFE module to enhance the spatial feature representation. To fully consider the self-similarity of pixels within the original HSI patch, two descriptors are constructed in the SSFE module. These two descriptors are fused with the original patch to exploit the original spatial location information effectively.

3) To reduce the computational complexity of self-attention, we propose two simple yet efficient modules, called SIT module and SCA module, for extracting local–global spatial–spectral information of HSIs. Moreover, they avoid excessive computational costs when applying self-attention on the network.

The rest of this article is organized as follows. Section II provides an overview of the related work. Section III presents our proposed network architecture. Section IV introduces the experimental results of the proposed method on four datasets. Finally, Section V concludes this article.

II. Related Work

In this section, we briefly introduce relevant works of HSIC, including CNN-based HSI classification, attention mechanism model, and transformer-based HSI classification.

A. CNN-Based HSI Classification

In recent years, CNNs have gained widespread attention in HSIC problems owing to their remarkable ability to extract local spectral-spatial features. Zhong et al. [10] proposed a spectral–spatial residual network, which incorporates 3-D CNN into spatial and spectral residual blocks in a residual connection manner to extract spatial–spectral information, respectively. [31] combines 3-D CNN with a boundary mirror strategy to achieve accurate classification results. Various works [32], [33] have combined 2-D CNN and 3-D CNN architectures to construct HSI classification models. He et al. [32] proposed a hybrid structure by integrating 2-D and 3-D CNNs. They further introduced a CNN dynamic accelerator structure tailored to this hybrid architecture, thus achieving more efficient classification. In [33], spatial shuffling operations were introduced into CNNs to simulate the real pixel position distribution, thereby solving the nonindependence problem between the training and test sets. Paoletti et al. [34] combined network architecture search with CNNs to automatically fine-tune networks in complex scenarios. Despite the commendable performance exhibited by CNN-based HSIC networks, their limited awareness of global information hinders further enhancement of classification accuracy.

B. Attention Mechanism Model

The attention mechanism can selectively focus on image regions that contain critical information by assigning different weights to various input components, thus effectively improving the performance of the algorithms. Hu et al. [35] proposed a plug-and-play squeeze-and-excitation module that focuses on channel attention, aiming to learn channel weights without additional computational costs. Based on channel attention, Woo et al. [36] integrated spatial attention into the network and verified that placing channel attention before spatial attention is more effective. Misra et al. [37] explored potential connections between spatial attention and channel attention, advocating for their simultaneous modeling. The triplet approach in this study models attention relationships from three perspectives: channel, height, and width. Cai et al. [38] further applied this idea to HSIC problems, implementing attention mechanisms across the vertical, horizontal, and spatial dimensions, thus achieving considerable results.

Self-attention is a variant of the attention mechanism that reduces the reliance on external information. Wang et al. [39] introduced nonlocal neural networks by incorporating a nonlocal operation to explore the relationships between positions. Huang et al. [40] proposed a more efficient attention mechanism, called criss-cross attention, to generate attention maps only using features along the cross path, thus significantly reducing computational complexity. ViT [20] first introduces transformer into computer vision tasks. Liu et al. [41] proposed a cascaded group attention module that applies segmented features to multihead attention. This approach effectively improves efficiency between attention heads. Zhang et al. [42] introduced the CNN-transformer blender module to fuse the features from CNN and transformer, thus achieving effective feature alignment. In addition, Hassani et al. [43] proposed neighborhood attention to calculate the attention between pixels and their nearest neighbors. Qiao et al. [44] further presented a multiscale neighborhood attention transformer network by incorporating neighborhood attention within a local window.

C. Transformer-Based HSI Classification

Transformer [45], originally introduced for natural language processing tasks, has demonstrated promising results for various vision tasks in the recent few years. Notably, many studies focus on exploring the application of transformer in HSIC tasks. Zhang et al. [46] developed a hybrid architecture by combining CNN with transformer, introducing two lightweight and efficient self-attention modules for HSIC. Ouyang et al. [47] introduced spectral–spatial attention instead of the self-attention module in the original transformer encoder, thereby effectively extracting long-range spectral–spatial information in HSI patches. Qi et al. [48] integrated 3-D CNN with transformer and then employed adaptive fusion to integrate multiscale features. Liao et al. [49] investigated long-range dependence across different spatial dimensions by pooling HSI patches with diverse scales, and then fed them into transformer through three separate branches. Thus, it can extract multiscale spatial information of HSIs. Furthermore, they designed a spectral transformer branch for the central pixel vector, followed by adaptive fusion among the branches. He et al. [50] leveraged interactive classification tokens to capture multiscale spectral information. Zhao et al. [51] combined transformer with graph CNNs to effectively learn node features and enhance classification accuracy. In [52], a center transformer was introduced to emphasize the spectral information at the center position, while a surrounding transformer was designed to dynamically interact with local fine- and coarse-grained information. Then, the features extracted from the center transformer and the surrounding transformer are combined for the final classification. Ma et al. [53] proposed a lightweight Gaussian self-attention to reduce the parameter number by enhancing the feature weight of the central pixel. Zhou et al. [54] developed a dual-branch approach for HSI

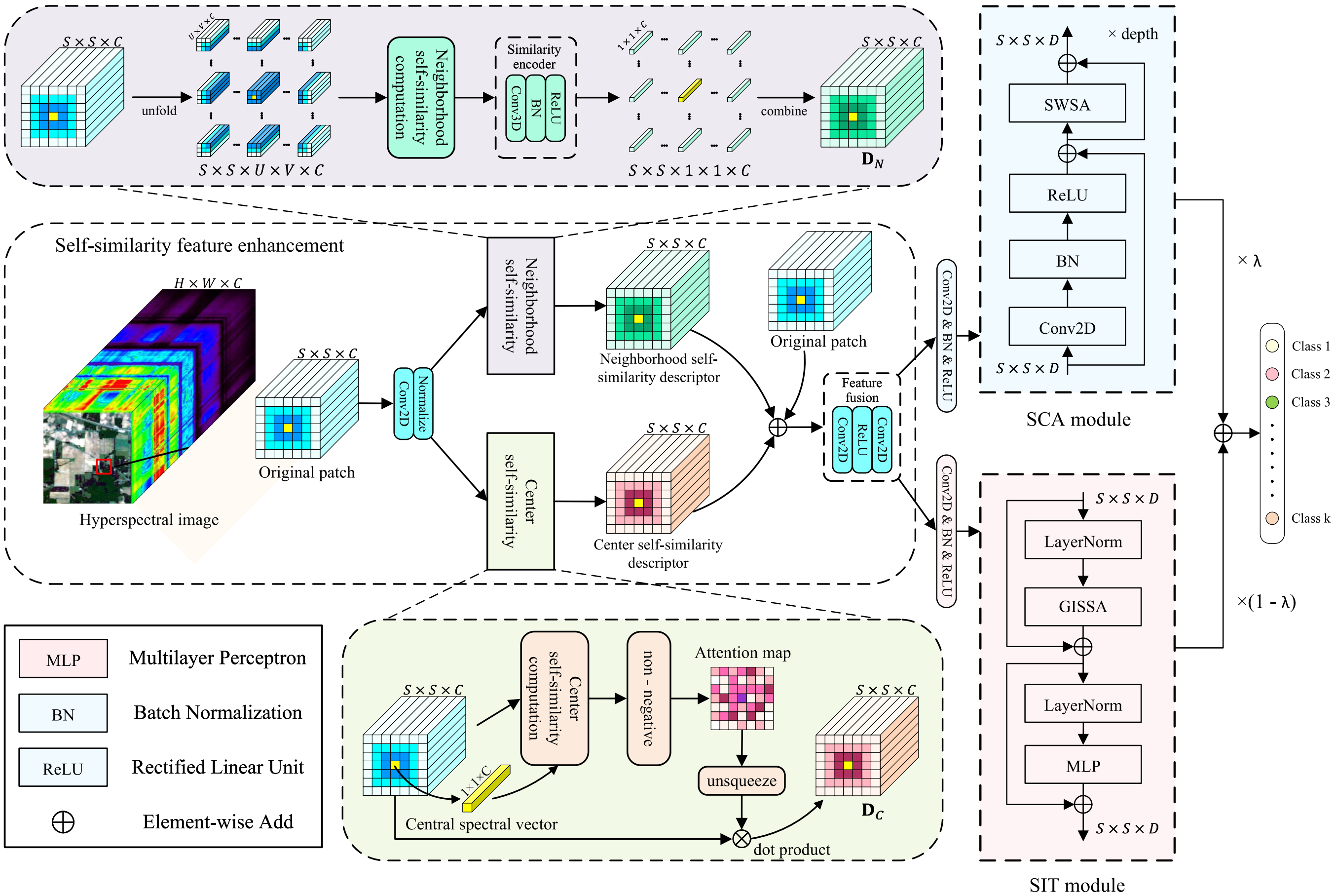

Fig. 1. Overview of the proposed EATN approach. It can be divided into a feature enhancement part and a spectral–spatial feature extraction part. The former aims to enhance the spatial feature representations of original patches. The latter consists of the SCA module for spatial information extraction and the SIT module for spectral information extraction. Then, we dynamically fuse the features of the two branches for the final classification.

classification tasks, in which the CNN branch employs multi-scale convolutions to extract local spatial–spectral information and the transformer branch performs self-attention across height, width, and spectral dimensions to achieve a deep fusion of spatial–spectral information.

III. Methodology

A. Overall Framework of the Proposed EATN Model

In this section, we provide a comprehensive description of the proposed method. Fig. 1 shows the overall framework of the proposed EATN method, which comprises three key components: the SSFE module, the SCA module, and the SIT module. Assume that the input HSIs can be expressed as $\mathbf{X} \in \mathbb{R}^{H\times W\times C}$, where H and W represent the length and width, respectively, and C denotes the number of bands. We take a selected pixel and its adjacent pixels as patch input to the proposed network. The patch can be expressed as $\mathbf{P} \in \mathbb{R}^{S\times S\times C}$, where $S \times S$ denotes the size of the patch. To enhance the spatial feature representation of the original patch, we feed the patch into the SSFE module, which generates two self-similarity descriptors with the same size and then fuses them with the original patch. Consequently,

TABLE I
Sample Division of the IP Dataset

No.	Name	Training	Validation	Test
1	Alfalfa	5	5	36
2	Corn–notill	25	25	1378
3	Corn–mintill	25	25	780
4	Corn	25	25	187
5	Grass–pasture	25	25	433
6	Grass–trees	25	25	680
7	Grass–pasture–mowed	5	5	18
8	Hay–windrowed	25	25	428
9	Oats	5	5	10
10	Soybean–notill	25	25	922
11	Soybean–mintill	25	25	2405
12	Soybean–clean	25	25	543
13	Wheat	25	25	155
14	Woods	25	25	1215
15	Building–grass–trees–drives	25	25	336
16	Stone–steel–towers	10	10	73
	Total	325	325	9599

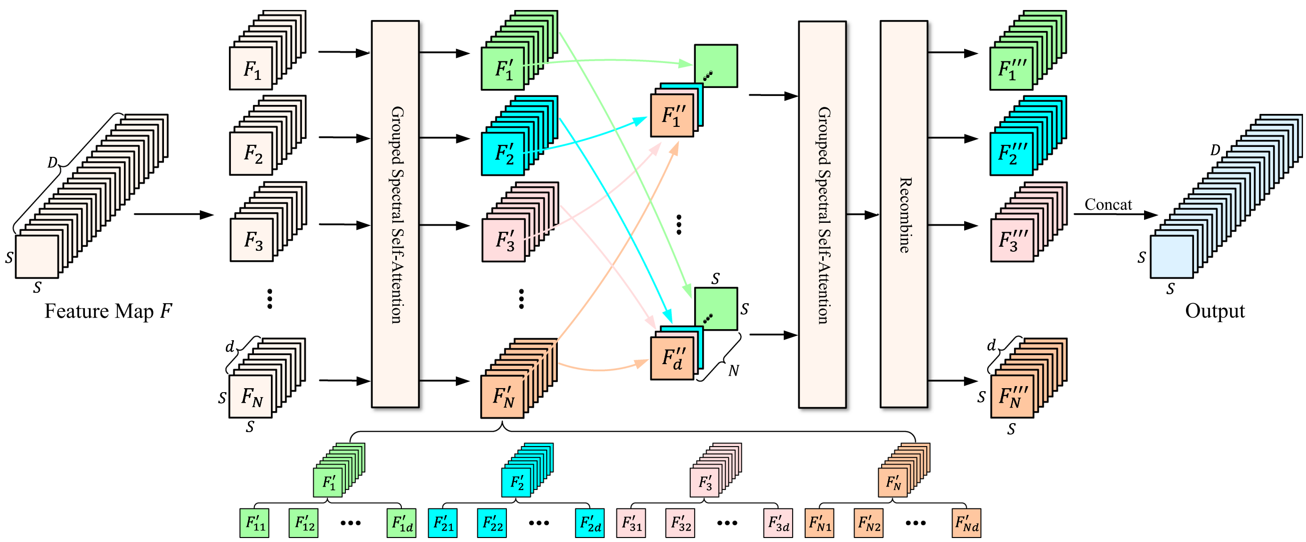

Fig. 2. Illustration of the proposed GISSA module. It implements two-step interactive self-attention calculations for reducing computational consumption.

TABLE II
SAMPLE DIVISION OF THE HU DATASET

No.	Name	Training	Validation	Test
1	Healthy grass	40	40	1171
2	Stressed grass	40	40	1174
3	Synthetic grass	10	10	677
4	Tree	40	40	1164
5	Soil	40	40	1162
6	Water	10	10	305
7	Residential	40	40	1188
8	Commercial	40	40	1164
9	Road	40	40	1172
10	Highway	40	40	1147
11	Railway	40	40	1155
12	Parking lot1	40	40	1153
13	Parking lot2	10	10	449
14	Tennis court	10	10	408
15	Running track	10	10	640
	Total	450	450	14129

TABLE III
SAMPLE DIVISION OF THE XZ DATASET

No.	Name	Training	Validation	Test
1	Bareland1	20	20	26356
2	Lakes	20	20	3987
3	Coals	20	20	2743
4	Cement	20	20	5174
5	Crops-1	20	20	13144
6	Tress	20	20	2396
7	Bareland2	20	20	6950
8	Crops	20	20	4737
9	Red-title	20	20	3030
	Total	180	180	68517

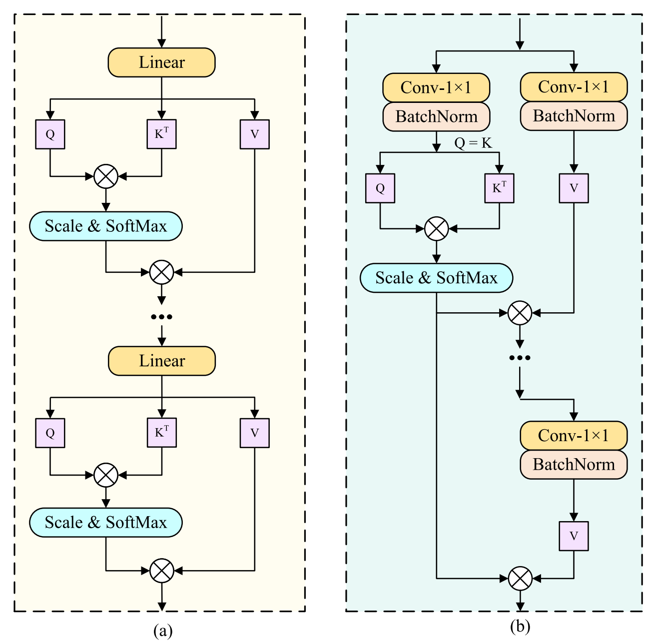

Fig. 3. Detail structure of the proposed SWSA module and original self-attention. (a) Original self-attention. (b) SWSA.

the size of the feature map remains $S \times S \times C$. Next, we input the feature map to the spatial and spectral feature extraction module. For the spectral branch, we employ a 2-D convolution with a kernel size of 1×1 to reduce the dimension of the feature map. Subsequently, we feed this into the proposed SIT module. For the spatial branch, we adopt a 2-D convolution with a kernel size of 3×3 to reduce the dimension of the feature map, and then feed it into a series of SCA modules. After dimensionality reduction, the dimension of the spatial and spectral branches is $S \times S \times D$. Furthermore, we initialize a trainable weighted parameter λ and perform a weighted sum of the feature maps

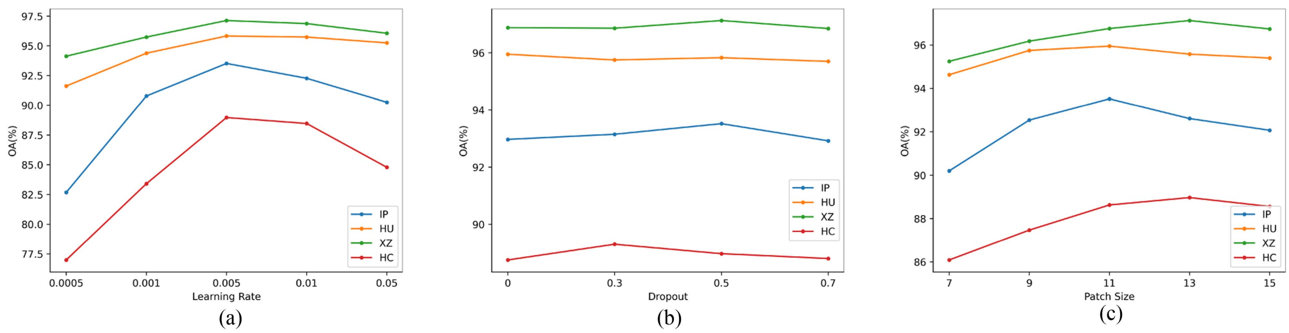

Fig. 4. Classification performance of the proposed EATN method under different conditions. (a) Learning rate. (b) Dropout. (c) Patch size.

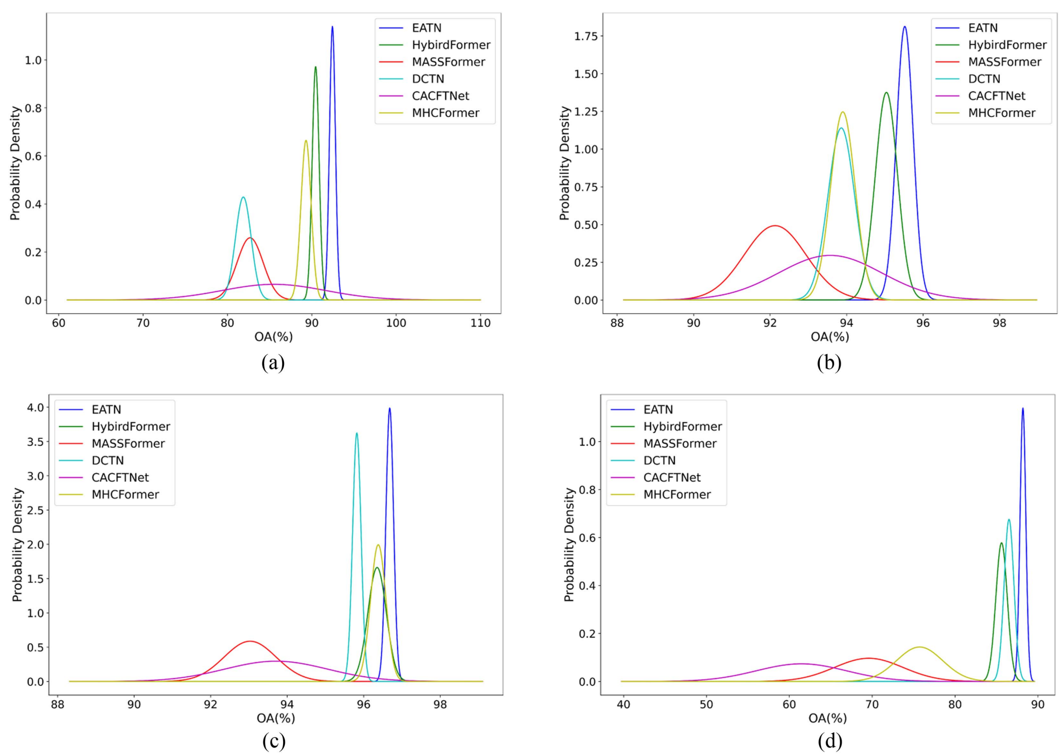

Fig. 5. Normal distribution curves for transformer-based HSI classification methods. (a) IP. (b) HU. (c) XZ. (d) HC.

from the two branches. Finally, we construct a linear classifier by integrating a dropout layer and a global average pooling, and then obtain the final classification results.

B. Self-Similarity Feature Enhancement

Due to the high-dimensional property of HSIs and the pattern of patch-based HSIC tasks, the relationships between pixels within the original input patch are crucial in HSIC problems. Since self-attention fails to effectively explore the local correlations, transformer-based models struggle to perceive the position information between pixels. Furthermore, it may be impractical to pursue a lightweight transformer model with a fast convergence rate and promising performance. To alleviate these issues, we design an SSFE module to process the original HSI patches.

Self-similarity signifies the similarity or repeatability between a specific local area in an image and either the whole image or other local areas. This similarity can manifest at various scales, ranging from microscopic pixel-level details to macroscopic overall structures. It plays a crucial role in image processing and computer vision. Several works utilize self-similarity to compute self-attention, assigning weights to image regions through an attention matrix. A self-similarity descriptor is designed to quantify and capture self-similarity characteristics in images or videos. These descriptors have been widely applied to computer vision, particularly in fields such as few-shot classification [55] and image matching [56]. In this work, we focus on developing self-similarity descriptors to enhance the feature information of HSIs.

In the SSFE module, we normalize the spectral dimension of the original patch to explore the relationships between pixels.

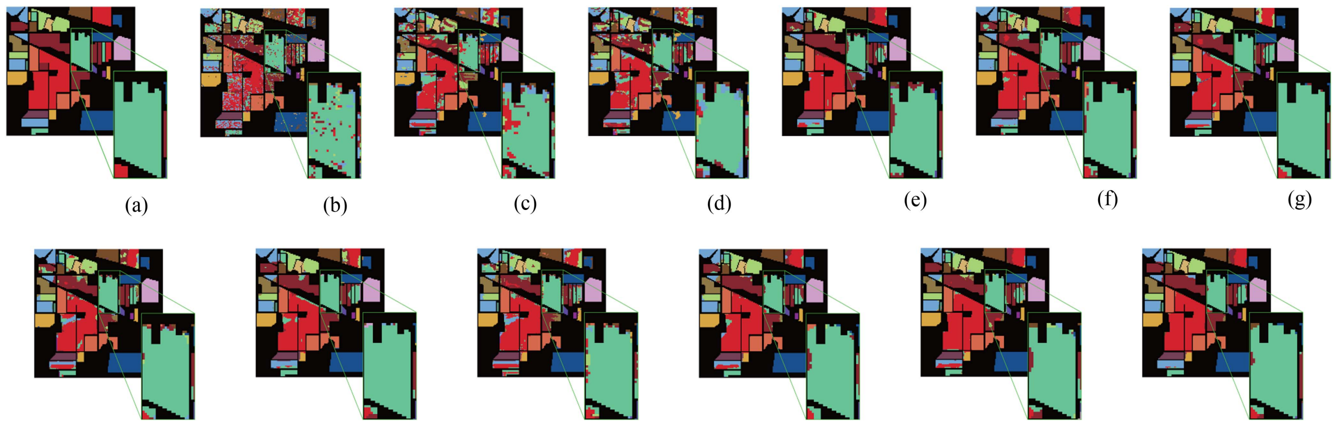

Fig. 6. Classification maps of different models on the IP dataset. (a) GT. (b) SVM. (c) 3D CNN. (d) RSSAN. (e) CLOLN. (f) S3ARN. (g) HSIC-SDM. (h) MASSFormer. (i) MHCFormer. (j) DCTN. (k) HybirdFormer. (l) CACFTNet. (m) Ours.

TABLE IV
SAMPLE DIVISION OF THE HC DATASET

No.	Name	Training	Validation	Test
1	Strawberry	44	44	44645
2	Cowpea	22	22	22707
3	Soybean	10	10	10266
4	Sorghum	5	5	5342
5	Water spinach	1	1	1197
6	Watermelon	4	4	4523
7	Greens	5	5	5891
8	Trees	17	17	17942
9	Grass	9	9	9450
10	Red roof	10	10	10494
11	Gray roof	16	16	16877
12	Plastic	3	3	3671
13	Bare soil	9	9	9097
14	Road	18	18	18522
15	Bright object	1	1	1133
16	Water	75	75	75250
	Total	249	249	257007

Subsequently, we define two types of self-similarity descriptors: neighborhood self-similarity descriptor and center self-similarity descriptor. These descriptors offer different perspectives on exploring the intrinsic relationships between pixels within the original patch. Both are then added to the original patch and fed into the feature-fusion module. The feature fusion module consists of two 2-D convolutional layers with 1×1 kernels and a ReLU layer. It effectively integrates the information from the original patch and the descriptors, thus enhancing the feature representation capability of the original patch. Consequently, the patch is more suitable as input for the proposed EATN method. Fig. 1 illustrates the SSFE module. In the following sections, we will introduce these two self-similarity descriptors.

1) Neighborhood Self-Similarity Descriptor: To preserve the local positional information of pixels within the original input patch, we introduce the self-similarity into the convolution operation, resulting in a novel neighborhood self-similarity descriptor. Specifically, assume that the input patch is $\mathbf{P} \in \mathbb{R}^{S\times S\times C}$, where $S \times S$ represents the size of the patch. We unfold the patch as a collection of pixels and their adjacent pixels, including overlapping regions. This can be represented as $\mathbb{R}^{S\times S\times U\times V\times C}$, where U and V denote the length and width of the selected neighborhood, respectively. Here, we pad the patch edges with zeros to facilitate the calculation of edge pixels. Next, the cosine self-similarity $R_{\text{Neighborhood}}$ for each pixel within its neighboring pixels with a $U \times V$ window is given as follows:

$$\mathbf{R}_{\text{Neighborhood}} = \frac{\mathbf{P}(\mathbf{x}, \mathbf{c}) \cdot \mathbf{P}(\mathbf{x}+\mathbf{z}, \mathbf{c})}{\|\mathbf{P}(\mathbf{x}, \mathbf{c})\| \|\mathbf{P}(\mathbf{x}+\mathbf{z}, \mathbf{c})\|} \tag{1}$$

where $\mathbf{z} \in [-d_U, d_U] \times [-d_V, d_V]$ corresponds to the relative position within the neighborhood window. Accordingly, $U = 2d_U + 1$ and $V = 2d_V + 1$, with U and V set to 3. $\mathbf{c}$ ranges from 1 to C, which represents the spectral band index in the pixel. We adopt channelwise calculation to generate the $R_{\text{Neighborhood}}$, aiming to preserve the semantic information from the original patch as much as possible. We subsequently apply nonnegative processing to the obtained self-correlation tensor.

To encode the self-correlation tensor from high-dimensional to compact descriptors, we feed the obtained self-correlation tensor into the similarity encoder block, which consists of a 3-D convolution layer, a batch normalization (BN) layer, and a ReLU layer. The number of similarity encoder blocks is related to the values of U and V. Subsequently, we reduce the spatial dimension of the self-correlation tensor from $U \times V$ to 1×1, thereby ensuring consistency in the size of the original input patch. Therefore, the neighborhood self-similarity descriptor $\mathbf{D}_N$ can effectively preserve local pixel relationship information, and enhance fundamental feature representations by merging it with the original patch.

2) Center Self-Similarity Descriptor: To highlight the significance of the center pixel in classification, we investigate the

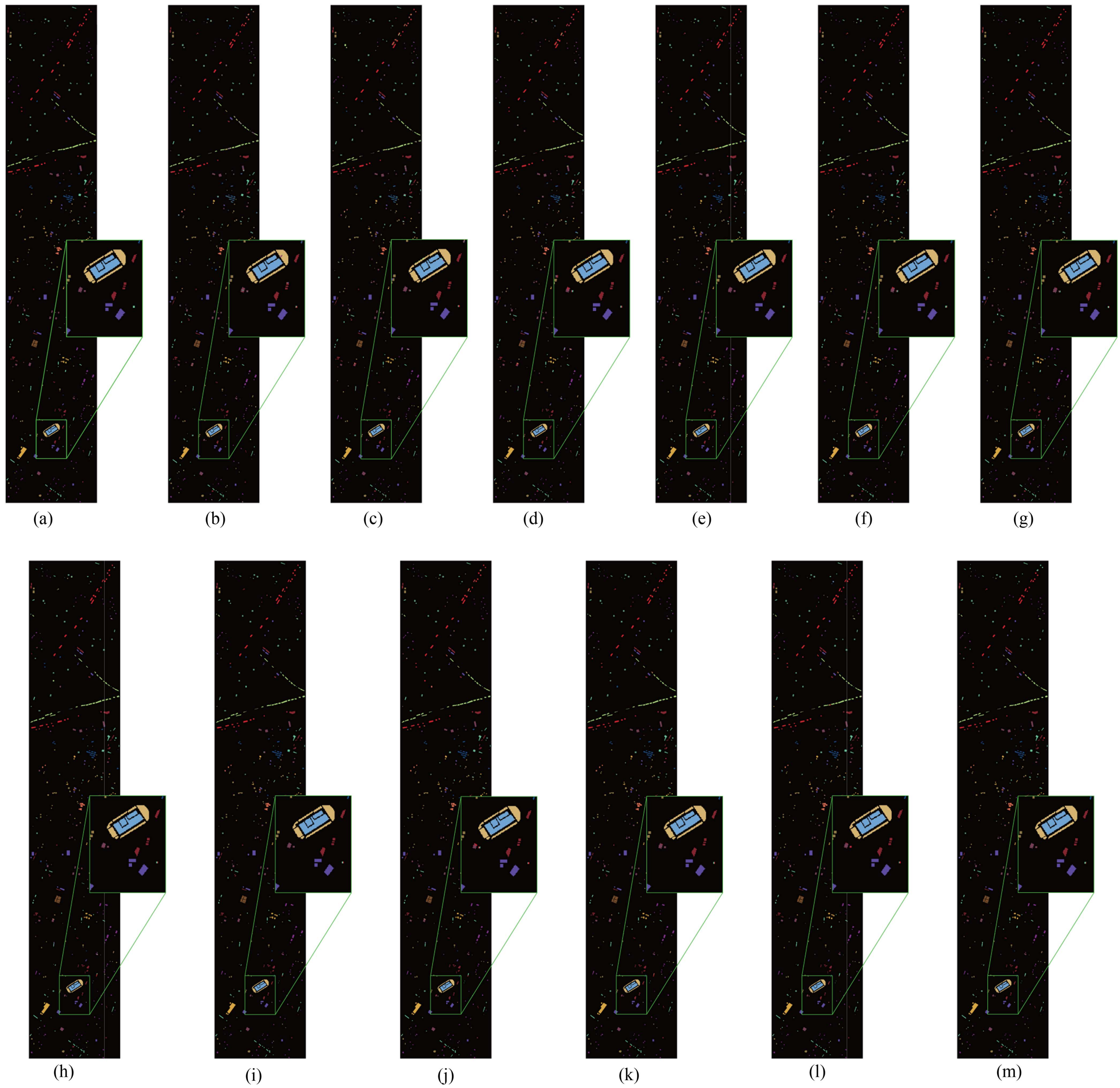

Fig. 7. Classification maps of different models on the HU dataset. (a) GT. (b) SVM. (c) 3-D CNN. (d) RSSAN. (e) CLOLN. (f) S3ARN. (g) HSIC-SDM. (h) MASSFormer. (i) MHCFormer. (j) DCTN. (k) HybirdFormer. (l) CACFTNet. (m) Ours.

relationship between the center pixel and other pixels within the input patch, and generate a novel center self-similarity descriptor. Considering the center pixel of the input patch as $\mathbf{x}_c \in \mathbb{R}^1 \times C$, we define the cosine self-similarity $\mathbf{R}_{\text{Center}}$ between the center pixel and all other pixels as follows:

$$\mathbf{R}_{\text{Center}} = \frac{\mathbf{P}(\mathbf{x}_c) \cdot \mathbf{P}(\mathbf{x}_c + \mathbf{z})}{\|\mathbf{P}(\mathbf{x}_c)\| \|\mathbf{P}(\mathbf{x}_c + \mathbf{z})\|} \quad (2)$$

where $\mathbf{z} \in [-d_S, d_S] \times [-d_S, d_S]$ and $S = 2d_S + 1$. The similarity map can be represented as $\mathbf{R}_{\text{Center}} \in \mathbb{R}^{S \times S \times 1}$. Similarly, we apply nonnegative processing to $\mathbf{R}_{\text{Center}}$, and then multiply $\mathbf{R}_{\text{Center}}$ with the original input patch $\mathbf{P} \in \mathbb{R}^{S \times S \times C}$ to derive the final center self-similarity descriptor $\mathbf{D}_C$. This descriptor effectively highlights pixel information that positively contributes to classification while mitigating the influence of irrelevant information.

Finally, we add the original input patch $\mathbf{P}$ to $\mathbf{D}_C$ and $\mathbf{D}_N$, and then input it to the feature fusion module. Therefore, we can get the feature map $\mathbf{F}$ of the feature fusion module as follows:

$$\mathbf{F} = \max\left(0, \left(\mathbf{P} + \mathbf{D}_C + \mathbf{D}_N\right) w_1 + b_1\right) w_2 + b_2 \quad (3)$$

where w_1, b_1, w_2, and b_2 represent the weights and biases of the first and second 2-D convolutional layers, respectively. We obtain the output feature map $\mathbf{F}$ from the SSFE module and pass it to the spatial–spectral feature extraction stage. Notably, the

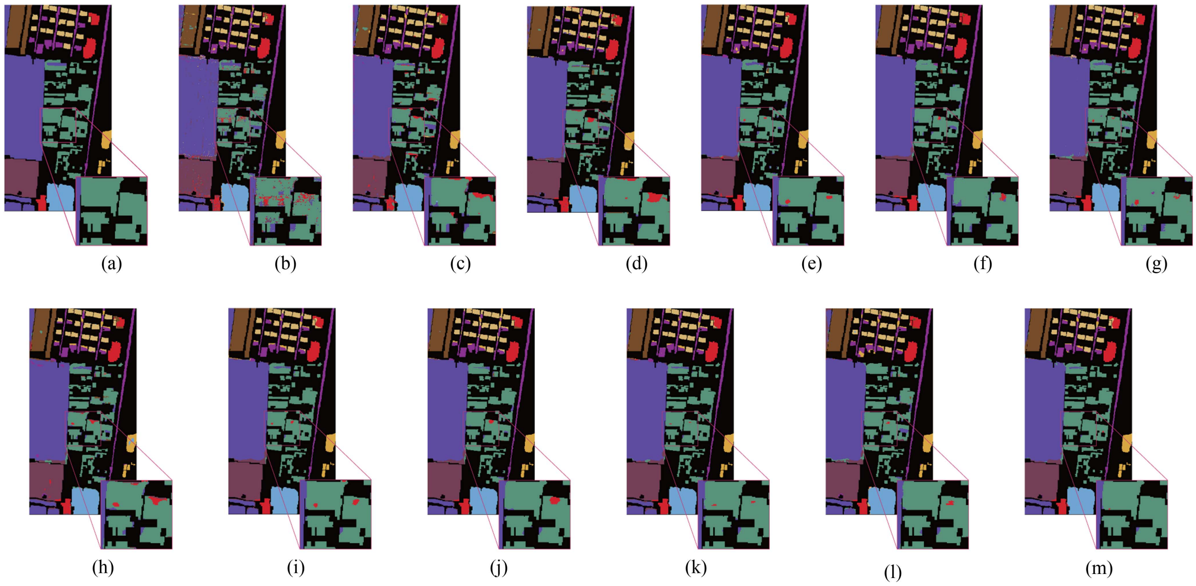

Fig. 8. Classification maps of different models on the XZ dataset. (a) GT. (b) SVM. (c) 3-D CNN. (d) RSSAN. (e) CLOLN. (f) S3ARN. (g) HSIC-SDM. (h) MASSFormer. (i) MHCFormer. (j) DCTN. (k) HybirdFormer. (l) CACFTNet. (m) Ours.

TABLE V
CLASSIFICATION RESULTS OF DIFFERENT MODELS ON THE HU DATASET

No.	SVM	3-D CNN	RSSAN	CLOLN	S3ARN	HSIC-SDM	MASSFormer	MHCFormer	DCTN	CACFTNet	HybirdFormer	Ours
1	95.42	**95.70**	92.87	93.45	95.56	94.19	94.04	93.55	95.54	95.07	94.39	95.15
2	96.81	96.01	89.21	95.47	97.07	94.65	96.72	94.42	95.51	89.62	**97.94**	96.51
3	99.69	100.00	97.38	100.00	100.00	100.00	100.00	99.69	100.00	96.34	100.00	**100.00**
4	97.99	98.85	98.05	98.20	97.81	97.54	99.02	97.88	97.48	96.49	98.94	**99.29**
5	93.85	96.38	93.18	**98.81**	94.01	96.32	94.29	97.27	95.38	96.48	97.44	97.51
6	97.77	86.69	65.52	85.76	83.81	86.56	92.96	82.28	81.99	96.36	92.53	**99.25**
7	89.59	91.95	90.65	95.39	96.49	96.09	91.60	**97.08**	95.83	96.33	95.88	96.95
8	86.91	83.80	86.65	95.22	**97.91**	93.98	92.62	94.75	97.42	95.08	95.75	95.26
9	88.63	86.80	87.59	93.38	96.97	96.43	93.27	93.73	95.99	**98.02**	95.44	92.15
10	91.10	89.53	86.35	91.70	94.03	94.69	**96.25**	96.15	88.55	91.06	89.77	92.59
11	82.70	76.21	87.28	96.86	94.53	**97.20**	92.80	97.20	94.36	95.51	96.61	96.36
12	87.99	74.71	70.92	92.79	83.02	92.37	86.42	88.33	92.79	**95.78**	94.35	94.77
13	49.90	90.27	81.08	88.91	87.83	91.59	76.45	86.30	86.12	92.60	90.85	**96.45**
14	93.37	81.56	79.81	90.64	92.93	83.79	94.83	89.46	98.38	95.31	97.87	**100.00**
15	**99.16**	97.99	93.94	98.04	95.23	98.52	93.46	96.29	93.73	95.08	95.52	94.17
OA (%)	90.62	89.26	87.51	94.88	94.24	95.02	93.33	94.49	94.48	94.87	95.64	**95.96**
AA (%)	90.05	89.76	86.69	94.30	93.81	94.26	92.98	93.62	93.93	95.00	95.55	**96.42**
KAPPA $\times$ 100	89.85	88.37	86.49	94.46	93.77	94.61	92.78	94.04	94.03	94.45	95.28	**95.63**

resulting feature map enhances the representation of features and highlights valuable pixel information compared to the original patch, thereby facilitating the feature extraction tasks.

C. SIT Module

Recently, many efforts [46], [47] have focused on using Transformer to model long-range dependencies between spectral bands. However, the transformer-based HSIC models need to consume expensive computational costs and huge storage space due to the high dimensionality of HSIs. Therefore, their applications are significantly limited in real HSIC tasks. In addition, spectral data typically exhibit continuity and local correlations, while self-attention in the transformer independently treats the features of each position in the sequence. Consequently, the transformer model fails to fully exploit the local correlations of spectral information.

To address the aforementioned challenges, we develop an SIT module by introducing the GISSA module, as illustrated in Fig. 2. The SIT module operates directly on the spectral dimension and conducts interactive self-attention calculations in groups. Specifically, for the input feature map $\mathbf{F} \in \mathbb{R}^{S \times S \times D}$, we divide it into N groups with equal length as follows:

$$\mathbf{F} = \{\mathbf{F}_1, \mathbf{F}_2, \ldots, \mathbf{F}_i, \ldots, \mathbf{F}_N\} \tag{4}$$

TABLE VI
CLASSIFICATION RESULTS OF DIFFERENT MODELS ON THE IP DATASET

No.	SVM	3-D CNN	RSSAN	CLOLN	S3ARN	HSIC-SDM	MASSFormer	MHCFormer	DCTN	CACFTNet	HybirdFormer	Ours
1	77.50	75.56	91.89	91.67	92.11	100.00	**100.00**	100.00	97.30	100.00	100.00	97.22
2	61.36	57.20	58.64	78.78	80.04	**93.11**	81.83	88.75	88.86	86.58	86.21	87.32
3	51.04	56.56	52.28	90.30	**92.21**	87.94	80.36	93.74	70.77	92.19	90.28	91.88
4	42.77	43.93	67.21	**96.30**	93.00	78.30	60.74	83.94	60.00	73.79	77.08	92.08
5	81.80	86.51	76.34	98.50	98.51	99.48	100.00	97.21	98.44	96.57	100.00	**100.00**
6	89.08	87.64	94.04	**99.40**	93.32	98.96	97.25	96.59	93.42	96.83	96.69	98.11
7	53.33	40.91	48.65	66.67	75.00	94.44	45.00	90.00	50.00	100.00	100.00	**100.00**
8	98.58	100.00	**100.00**	96.15	94.63	98.38	99.53	93.83	96.15	95.51	100.00	99.07
9	16.00	45.45	32.26	32.26	40.00	58.82	52.63	45.45	83.33	58.33	**76.92**	47.62
10	64.05	55.55	59.73	83.71	71.96	72.75	74.26	73.42	67.09	83.88	79.91	**87.09**
11	82.60	75.34	84.71	92.40	93.11	90.59	88.61	94.69	85.85	90.27	91.81	**95.68**
12	60.96	48.55	42.73	89.01	96.69	79.06	75.29	83.51	71.45	83.83	93.14	**99.21**
13	89.47	87.01	74.75	93.87	86.86	**95.00**	79.38	79.58	82.61	93.25	91.67	93.87
14	97.61	93.47	92.67	97.31	98.76	98.93	97.14	99.25	98.17	**99.75**	99.34	98.36
15	51.15	63.81	78.95	88.59	86.06	**99.10**	85.98	92.96	86.22	87.00	88.95	85.57
16	79.35	**89.47**	84.71	83.54	82.95	84.71	74.49	64.15	68.22	78.89	77.42	81.82
OA (%)	72.07	70.71	72.73	90.15	88.94	90.06	85.74	90.09	83.71	90.41	91.00	**93.52**
AA (%)	68.54	69.18	71.22	86.15	85.95	89.34	80.78	86.06	81.11	88.54	90.58	**90.93**
KAPPA×100	68.51	66.75	69.09	88.73	87.37	88.66	83.78	88.70	81.44	89.02	89.70	**92.58**

TABLE VII
CLASSIFICATION RESULTS OF DIFFERENT MODELS ON THE XZ DATASET

No.	SVM	3-D CNN	RSSAN	CLOLN	S3ARN	HSIC-SDM	MASSFormer	MHCFormer	DCTN	CACFTNet	HybirdFormer	Ours
1	94.25	95.00	96.17	98.31	98.05	98.70	97.48	99.22	98.65	97.52	99.07	**99.46**
2	99.87	99.09	99.92	99.80	100.00	99.70	97.15	99.87	99.87	99.69	**100.00**	99.64
3	79.34	82.54	87.89	89.15	**93.44**	86.29	84.19	89.37	87.65	79.49	91.34	87.98
4	92.60	82.26	95.06	96.41	96.18	96.97	90.03	97.20	97.59	96.34	97.28	**98.84**
5	97.50	95.11	96.90	98.12	96.77	96.54	95.67	**98.24**	97.15	94.16	97.41	96.26
6	70.32	74.61	85.04	93.27	89.41	94.08	85.81	94.79	93.98	92.93	94.89	**95.84**
7	72.83	84.71	87.71	91.37	88.57	87.63	88.51	88.95	90.56	86.52	89.19	**93.98**
8	96.03	94.15	83.64	96.57	95.19	96.80	89.59	**98.46**	95.03	93.24	96.61	97.12
9	93.20	84.01	80.18	95.56	96.06	88.76	97.01	91.26	88.76	**96.33**	93.95	93.75
OA (%)	90.59	91.19	92.94	96.61	95.90	95.59	93.92	96.73	96.04	94.21	96.65	**97.13**
AA (%)	88.43	87.94	90.27	95.39	94.85	93.94	91.71	95.26	94.35	92.91	95.52	**95.87**
KAPPA×100	88.13	88.85	91.06	95.72	94.81	94.44	92.32	95.87	95.00	92.69	95.77	**96.37**

where $\mathbf{F}_i \in \mathbb{R}^{S\times S\times d}$, where $d = D/N, N$ set to 16. S represents the size of the input feature map. For each $\mathbf{F}_i$, we consider its channels as tokens and perform self-attention operations to capture fine-grained interspectral information. To this end, we employ a 1×1 convolution layer to map $\mathbf{F}_i$ to the corresponding $\mathbf{Q}_i$, $\mathbf{K}_i$, and $\mathbf{V}_i$. By taking the dot product of the transposed $\mathbf{Q}_i$ and $\mathbf{K}_i$, we obtain the attention map $\mathbf{M}_i$. To mitigate the vanishing gradient problem, we divide the feature map by $\sqrt{d_k}$ as follows:

$$\mathbf{M}_i = \frac{\mathbf{Q}_i {\mathbf{K}_i}^T}{\sqrt{d_k}} \tag{5}$$

where $\mathbf{Q}_i$, $\mathbf{K}_i \in \mathbb{R}^{d\times n}$, where $n = S \times S$. Subsequently, we apply the signed square root operation to the attention map $\mathbf{M}_i$ and introduce a fixed bias δ. Afterward, we normalize the map using the softmax function. Then, the attention map is multiplied by $\mathbf{V}_i$ to obtain the output $\mathbf{F}'_i$ of the first self-attention as follows:

$$\mathbf{F}'_i = \text{softmax}\left(\text{sign}(\mathbf{M}_i) \cdot \sqrt{|\mathbf{M}_i| + \delta}\right) \mathbf{V}_i. \tag{6}$$

In addition, we incorporate a residual connection to fuse $\mathbf{F}'_i$ and $\mathbf{F}$. After the first self-attention calculation, we apply BN and ReLU activation to the resulting $\mathbf{F}'_i$. Then, we perform the second self-attention calculation on the processed feature $\mathbf{F}'_i \in \mathbb{R}^{S\times S\times d}$. To achieve global spectral information interaction, we reorganize channels with the same index from different $\mathbf{F}'_i$ instances into a new $\mathbf{F}''_i$, serving as input for the second self-attention calculation. Let $\mathbf{F}'_i = \{\mathbf{F}'_{i1}, \mathbf{F}'_{i2}, \ldots, \mathbf{F}'_{id}\}$ represent the original set of $\mathbf{F}'_i$ instances, where $\mathbf{F}'_{ij}$ represents the jth channel in the $\mathbf{F}'_i$. The reorganized feature set $\mathbf{F}''_i$ can

TABLE VIII
CLASSIFICATION RESULTS OF DIFFERENT MODELS ON THE HC DATASET

No.	SVM	3-D CNN	RSSAN	CLOLN	S3ARN	HSIC-SDM	MASSFormer	MHCFormer	DCTN	CACFTNet	HybirdFormer	Ours
1	85.14	76.67	79.97	88.76	86.06	87.74	82.57	82.80	89.16	77.17	90.56	**94.27**
2	56.90	63.04	66.60	74.90	78.13	78.27	73.74	73.77	**86.49**	56.20	74.20	78.97
3	61.47	34.67	36.73	51.58	59.61	55.00	51.72	58.66	71.34	61.57	77.49	**82.66**
4	84.23	35.55	35.72	79.63	90.31	68.35	68.70	80.78	**97.75**	00.00	93.30	94.49
5	21.88	16.02	3.92	00.00	75.00	53.33	95.83	00.00	80.45	00.00	77.12	**100.00**
6	17.82	17.18	11.21	6.41	24.52	22.69	54.98	41.86	**64.89**	00.00	42.61	45.14
7	55.34	59.09	67.87	58.24	**88.40**	39.83	40.26	57.21	79.78	00.00	82.86	84.99
8	67.92	52.50	41.87	66.79	55.93	68.08	57.43	60.23	70.61	56.31	71.99	**81.60**
9	52.56	53.32	50.09	60.70	50.46	**82.82**	82.71	78.20	69.58	47.37	71.50	65.41
10	86.81	65.03	71.44	70.52	82.41	88.93	93.02	**99.20**	95.38	73.76	92.24	98.82
11	82.95	84.93	90.15	89.49	92.66	84.59	83.65	69.23	95.00	73.85	**97.35**	90.96
12	40.77	12.15	26.54	00.00	46.03	33.75	00.00	00.00	**90.28**	00.00	79.05	72.99
13	39.02	48.58	45.60	36.39	55.23	36.96	33.37	35.39	57.45	26.34	53.04	**67.44**
14	78.28	79.23	77.85	76.10	81.22	76.69	74.09	78.68	**84.94**	64.27	80.01	83.56
15	97.31	99.09	9.09	96.19	99.64	88.86	**100.00**	100.00	87.30	00.00	100.00	97.12
16	95.53	94.98	97.06	99.23	99.87	97.29	93.21	96.72	99.31	**99.92**	98.92	99.89
OA (%)	77.67	73.66	74.44	80.95	82.55	80.20	77.22	78.64	87.97	74.66	86.52	**89.31**
AA (%)	63.99	55.75	50.73	59.68	72.84	66.44	67.82	63.29	82.48	39.79	80.13	**83.64**
KAPPA×100	73.75	68.89	69.91	77.60	79.51	76.71	73.04	74.78	85.86	69.93	84.15	**87.47**

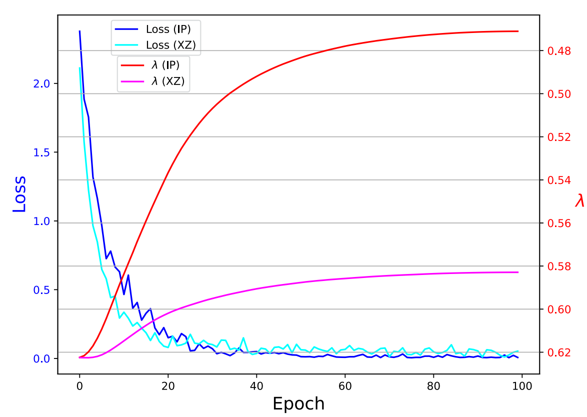

Fig. 9. Loss value of the proposed model with different settings of the balance operator λ.

be expressed as follows:

$$\mathbf{F}_i'' = \{\mathbf{F}_{1i}', \mathbf{F}_{2i}', \ldots, \mathbf{F}_{Ni}'\}, i = \{1, 2, \ldots, d\}. \quad (7)$$

We compute the reorganized feature using the same method as the first self-attention calculation. Let $\mathbf{F}_i''' = \{\mathbf{F}_{1i}''', \mathbf{F}_{2i}''', \ldots, \mathbf{F}_{Ni}'''\}$ represent the feature set after the second self-attention calculation. We restore the channels to their initial order based on their respective indices, which can be expressed as follows:

$$\mathbf{F}_i'''' = \{\mathbf{F}_{i1}''', \mathbf{F}_{i2}''', \ldots, \mathbf{F}_{id}'''\}, i = \{1, 2, \ldots, N\}. \quad (8)$$

Then, we concatenate the restored feature $\mathbf{F}_i''''$ to produce the final output of GISSA. Thus, the total self-attention computing can be given as follows:

$$\text{Attention}(\mathbf{F}) = \text{Concat}(\mathbf{F}_1'''', \mathbf{F}_2'''', \ldots, \mathbf{F}_N''''). \quad (9)$$

TABLE IX
PARAMS, FLOPS, AND INFER OF TRANSFORMER-BASED HSIC METHODS ON THE IP DATASET

Method	Params (M)	FLOPs (G)	Infer (s)
CACFTNet	<u>1.9300</u>	14.8973	1.1773
MHCFormer	3.8116	29.8735	1.0488
DCTN	45.2797	48.0034	2.9729
HybirdFormer	3.3686	<u>10.4179</u>	<u>0.5877</u>
MASSFormer	3.3962	10.8646	1.2197
Ours	**1.2607**	**9.7501**	**0.5471**

TABLE X
ABLATION RESULTS OF THE PROPOSED MODEL ON FOUR DATASETS

Case	Component				OA (%)			
	SIT	SCA	SSFE(N)	SSFE(C)	IP	HU	XZ	HC
1	×	×	✓	✓	82.14	90.36	91.86	67.27
2	✓	×	✓	✓	92.81	95.34	96.92	87.93
3	×	✓	✓	✓	92.66	95.54	96.44	88.41
4	–	✓	✓	✓	92.92	95.64	96.81	88.74
5	✓	–	✓	✓	92.77	95.44	96.97	89.15
6	✓	✓	×	×	89.86	94.21	95.32	81.65
7	✓	✓	✓	×	92.85	95.66	97.01	88.69
8	✓	✓	✓	✓	**93.52**	**95.95**	**97.13**	**89.30**

Inspired by Vaswani et al. [45], we introduce a linear normalization layer before the attention layer and the MLP layer, and incorporate residual connections within the layers, simultaneously. MLP is expressed as follows:

$$\text{MLP}(\mathbf{X}) = \text{FC}_2\left(\text{GeLU}\left(\text{FC}_1(\mathbf{X})\right)\right) \quad (10)$$

where $\text{FC}_i(.)$ denotes the ith fully connected layer and GeLU represents the Gaussian error linear unit. The SIT module is

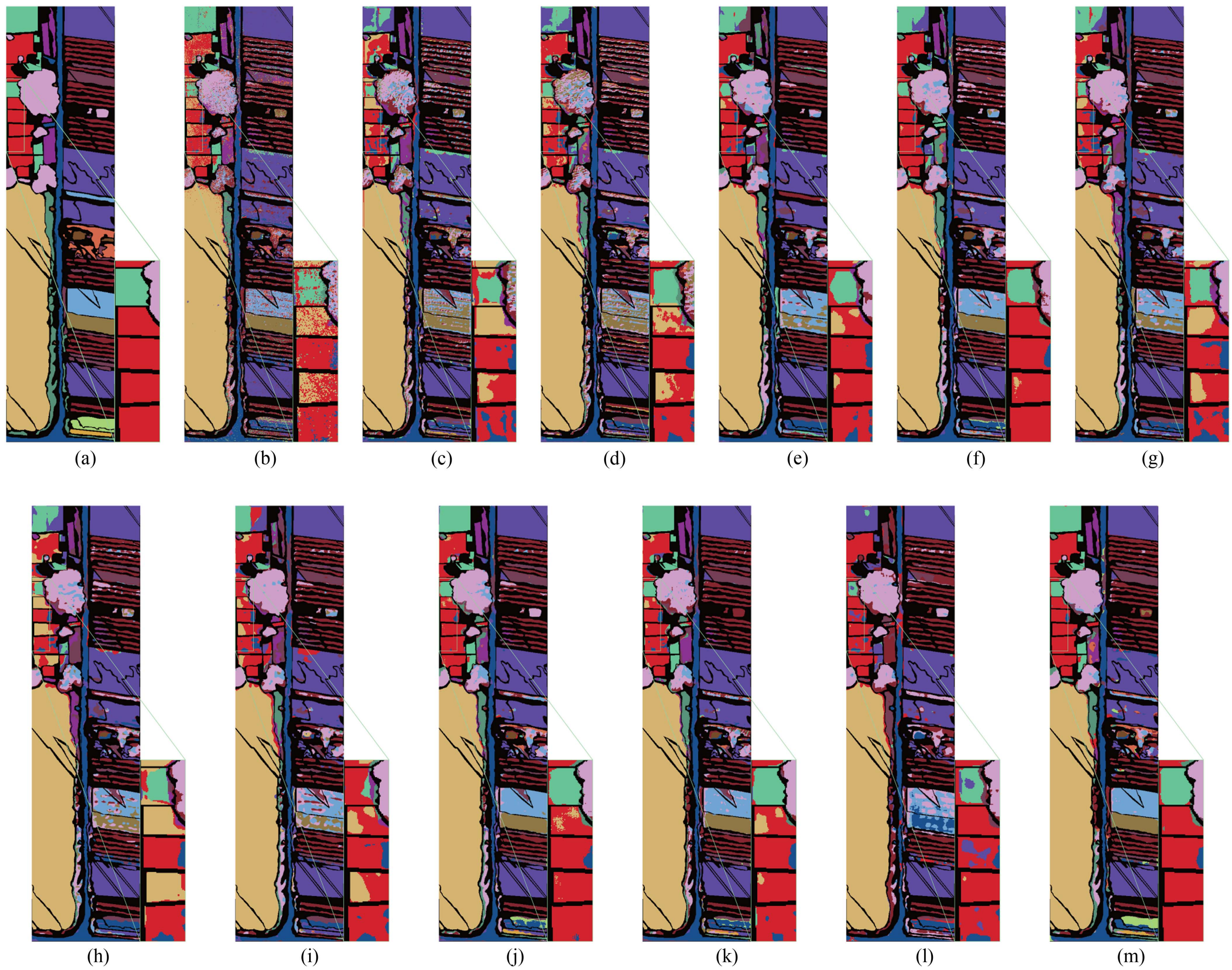

Fig. 10. Classification maps of different models on the HC dataset. (a) GT. (b) SVM. (c) 3D CNN. (d) RSSAN. (e) CLOLN. (f) S3ARN. (g) HSIC-SDM. (h) MASSFormer. (i) MHCFormer. (j) DCTN. (k) HybirdFormer. (l) CACFTNet. (m) Ours.

mathematically expressed as follows:

$$\mathbf{A} = \text{Attention}\big(\text{LayerNorm}(\mathbf{X})\big) + \mathbf{X} \tag{11}$$

$$\mathbf{O} = \text{MLP}\left(\text{LyerNorm}\left(\mathbf{A}\right)\right) + \mathbf{A} \tag{12}$$

where $\mathbf{A}$ and $\mathbf{O}$ denote the attention layer and the MLP layer output features, respectively. By employing a two-step interactive self-attention calculation, IST effectively captures local fine-grained spectral information while facilitating the interaction of global information. Moreover, we can obtain these results without additional computational costs.

D. SCA Module

To effectively extract discriminative local–global spatial features of HSIs, we propose an SCA module, which consists of a convolution sequence and a self-attention layer. The convolution sequence consists of a 3×3 convolution layer, a BN layer, and a ReLU layer. It aims to explore local information by leveraging the inductive bias built-in the convolution. Simultaneously, we establish global dependencies through an improved self-attention layer. Specifically, we replace fully connected layers with 1x1 convolutional layers for input feature mapping [57]. In addition, we only use two independent 1×1 convolutions to map the input features $\mathbf{F}$ to $\mathbf{Q}$ and $\mathbf{V}$, and set $\mathbf{Q} = \mathbf{K}$, which eliminates one convolution in each layer [58]. In contrast to the widely used LayerNorm that divides the calculation of self-attention into numerous element-level operations [59], [60], we adopt BN after each convolution to ensure stable training. BN is integrated into the convolution operation without introducing additional computational overhead.

Moreover, the attention maps in self-attention involve extensive elementwise calculations, and stacking of the self-attention layers leads to a substantial computational burden. To address this issue, we introduce shared weight self-attention (SWSA) into the SCA modules. After calculating the first attention map in self-attention, we reuse it in the following several SCA modules without compromising model performance, as illustrated in Fig. 3. This approach significantly reduces computational costs for subsequent attention map generation and convolution operations.

In addition, we add a residual connection between the convolutional sequence and the self-attention layer. By alternately using the convolutional sequence and the self-attention layer, the module outputs the feature map $\mathbf{F}_{\text{spatial}}$ containing local–global spatial information. We combine the feature map $\mathbf{F}_{\text{spectral}}$ extracted from the SIT module with the feature map $\mathbf{F}_{\text{spatial}}$ through dynamic integration as follows:

$$\mathbf{F} = (\lambda \times \mathbf{F}_{\text{spatial}} + (1 - \lambda) \times \mathbf{F}_{\text{spectral}}) \tag{13}$$

where λ is a learnable parameter.

IV. Experiment and Analysis

In this section, we conducted some experiments on four benchmark HSI datasets to assess the effectiveness of our proposed model. To assess the effectiveness of our EATN method, we compared it with several state-of-the-art CNN-based and transformer-based HSIC methods.

A. Datasets Description

1) Indian Pines (IP): The IP dataset was captured in Indiana, United States for early HSIC problems. It was obtained using the airborne visible/infrared imaging spectrometer, which operates in the wavelength range of 400–2500 nm. The dataset consists of a cropped area measuring 145 × 145 and the spatial resolution is approximately 20 m. 20 bands (104–105, 150–163, 220) related to water absorption are excluded from this dataset. Thus, 200 bands were utilized for training. The study area covers 16 land cover categories. The image consists of a total of 21 025 pixels, with 10 249 classified as object pixels and 10 776 as background pixels.

2) Houston 2013 (HU): The HU dataset, comprising 144 bands, was acquired in HU, Texas, USA, along with the surrounding rural areas. It contains 349 × 1905 pixels and covers a 2.5-m spatial resolution. The band range of this dataset spans from 364 to 1046 nm. The study area consists of 15 different land cover categories, including roads, soil, trees, highways, etc.

3) Xuzhou (XZ): The XZ dataset, collected in November 2014, consists of 500 × 260 pixels captured by an airborne HYSPEX hyperspectral camera over a periurban site in XZ. It offers a high spatial resolution of 0.73 m/pixel and uses 436 spectral bands, with noise bands removed from 415 to 2508 nm. The scene includes nine categories, such as crops, vegetation, manmade structures, and coal fields. The high resolution and diverse categories pose serious challenges to classification.

4) WHU-Hi-HanChuan (HC): The HC dataset was collected on 17 June 2016, from 17:57 to 18:46 in Hanchuan, Hubei Province, China. During the data acquisition, the temperature is approximately 30°C and the relative humidity is around 70%. The UAV operated at a flight altitude of 250 m, capturing images with dimensions of 1217 × 303 pixels. These pixels correspond to 16 types of ground objects, including strawberries, cowpeas, soybeans, sorghum, water spinach, watermelons, etc. The HSIs cover 274 bands within the 400–1000 nm range, with a spatial resolution of approximately 0.109 m.

B. Experimental Settings

To evaluate the performance of the proposed HSIC method, we utilized three metrics: overall accuracy (OA), average accuracy (AA), and Kappa coefficient (KAPPA). Generally speaking, a higher value indicates better classification results. The labeled pixels from the datasets were randomly divided into training, validation, and testing sets. Training sample proportions were determined based on dataset size. Tables I–IV present the number of pixels in the training, validation, and testing sets of each HSI dataset, respectively. To mitigate the issue of overlap between the training and testing sets, smaller training and validation sets were usually configured, and more pixels were allocated for testing. Approximately 3%, 3%, 0.25%, and 0.1% samples were selected as training sets for IP, HU, XZ, and HC datasets, respectively. We set the same number of samples in both the training and validation sets for each dataset.

To ensure a fair comparison, each HSIC method was run with the same configuration. All methods were trained independently for 100 epochs. The model parameters that yielded the best OA on the validation set were chosen for testing, and the results obtained on the test set were considered as the final results. In these experiments, each HSIC method ran ten times and then selected the best OA as the model performance.

The virtual environment was configured with Python 3.9.17, PyTorch 2.0.0, and CUDA 11.8. The hardware setup consists of an NVIDIA GeForce RTX 3090 GPU with 24 GB of memory and an Intel(R) Core(TM) i9-12900KF CPU.

C. Experiment Parameter Tuning

Hyperparameter settings directly influence the classification results of our model, and thus fine-tuning these hyperparameters can further enhance model performance. In this section, we systematically evaluated the impact of hyperparameters on the effectiveness and efficiency of our model.

Learning rate is a crucial hyperparameter for training deep learning models. Setting the learning rate too high or too low can hinder the convergence rate or the performance. To investigate the impact of different learning rates on the proposed method, we conducted experiments on four datasets and analyzed their performances, as shown in Fig. 4(a). The results indicate that different learning rates significantly impact the classification performance of the model. Due to limited training samples, the proposed method exhibits a certain sensitivity to the learning rate on the IP and HC datasets. Notably, the best performances on the IP, XZ, and HC datasets are obtained by setting the learning rate to 0.005. The best performance on the HU dataset can be achieved at a learning rate of 0.1 and 0.005. For the sake of consistency, we set the learning rate to 0.005 in the following experiments on all four datasets.

Dropout, as a regularization technique commonly employed in deep learning models, can effectively avoid overfitting. In the proposed EATN method, we incorporated dropout before network classification to mitigate the overfitting problem. Fig. 4(b) presents the classification performance of the proposed method on the IP, HU, XZ, and HC datasets with different dropout settings. The proposed method achieves superior classification performance with a dropout rate of 0.5 on the IP and XZ datasets. In addition, the best results are achieved by setting the dropout rate to 0 and 0.3 on the HU dataset and HC dataset, respectively. Consequently, we chose the optimal dropout rate for different datasets.

To evaluate the influence of different patch sizes on classification performance, the patch sizes ranged from 7×7 to 15×15 on four datasets. Fig. 4(c) shows the performances of the proposed method varied the path size. It can be seen that a smaller patch size, such as 7×7 or 9×9, results in relatively poor performance. In general, as the patch size increases, the classification performance also increases correspondingly. This can be attributed to the fact that a larger patch size contains more discriminative information. However, an excessively large patch size may introduce more noisy pixels, thereby potentially affecting overall classification results. We can observe that the optimal OA value for the IP and HU datasets is achieved by setting the patch size to 11×11, whereas the optimal value for the XZ and HC datasets is attained with a patch size of 13×13.

D. Experimental Results

Tables VI–VIII present the classification results of the proposed method and several state-of-the-art methods on the four datasets. Several HSIC methods including traditional classifiers (SVM) [5], CNN-based methods (3-D CNN [61], RSSAN [26], CLOLN [12], S3ARN [27], and HSIC-SDM [62]), as well as transformer-based methods (MASSFormer [59], MHCFormer [60], DCTN [54], CACFTNet [63], and HybirdFormer [47]), are used to evaluate the effectiveness of the proposed EATN method.

1) Quantitative Evaluation: The SVM with a radial basis function has demonstrated stable and significant results on four datasets. However, it ignores the spatial information of HSIs, and thus fails to extract discriminative information from the pixels fully. 3-D CNN achieves the lowest OA with values of 70.71% and 73.66% on the IP and HC datasets, respectively. This can be attributed to the fact that this simple network structure cannot effectively integrate shallow and deep information. Although the RSSAN method improves classification performance by combining spectral–spatial attention with 3-D CNN, it still achieves unsatisfactory performance. Moreover, the HSIC-SDM method performs better than RSSAN on four datasets by designing two branches to refine the spatial and spectral information extracted from the feature map.

It can be observed that the proposed method outperforms other state-of-the-art HSIC methods on four datasets. Despite the discrete characteristics exhibited by most classifiable pixels in the HU dataset, the proposed method still achieves an OA of 95.96% on this dataset. In comparison to HybirdFormer, CACFTNet, DCTN, MHCFormer, MASSFormer, HSIC-SDM, S3ARN, CLOLN, RSSAN, 3-D CNN, and SVM, the proposed method improves OA by 0.32%, 1.09%, 1.48%, 1.47%, 2.63%, 0.94%, 1.72%, 1.08%, 8.45%, 6.7%, and 5.34%, respectively. We can find that both MASSFormer and CACFTNet exhibit relatively lower accuracy compared to the proposed method. The main reason is that they primarily establish global dependencies in the spatial dimension while ignoring the spectral global information. In contrast, the proposed EATN considers global spectral–spatial information, thus achieving significant performance improvement. It can be seen that the HybridFormer achieved an OA, AA, and Kappa of 95.64%, 95.55%, and 95.28%, respectively, showing a decrease of 0.32%, 0.87%, and 0.35% compared to our EATN method. It can be attributed to the proposed EATN method explores the pixel relationships within the input original patches, thus effectively preserving their spatial information.

It can be seen that several 3-D CNN-based methods (3D CNN, RSSAN) and transformer-based methods (DCTN) exhibit a significant decline in classification accuracy on the IP datasets. The main reason for this phenomenon is that they cannot effectively train the 3-D CNN-based networks with limited training samples, resulting in network overfitting. Although DCTN performs well on other datasets, it achieves only 83.71%, 81.11%, and 81.44% OA, AA, and KAPPA on the IP dataset, respectively. Nevertheless, the proposed method still achieves the best classification results, highlighting the effectiveness of our feature enhancement strategy. Meanwhile, it also indicates that our proposed method can utilize feature information more efficiently with limited samples. We can see that all HSIC models show promising classification performance on the HU and XZ datasets with sufficient training samples. However, our proposed EATN model obtains the best classification results, showcasing its effectiveness in HSIC tasks.

2) Visual Evaluation: Figs. 6–10 visually illustrate the classification results of different methods on four HSI datasets. It can be found that both SVM and 3-D CNN exhibit a notable number of misclassified pixels. In contrast to the relatively continuous misclassified pixels observed in 3-D CNN, the classification map of SVM displays discrete salt and pepper noise. This is because SVM fails to consider the spatial neighborhood information of HSIs. Due to the nonsmooth distribution of pixels in the IP dataset, many approaches struggle to perform well in certain categories, such as Soybean–notill. Moreover, our proposed method consistently achieves considerable classification performance in such scenarios. Overall, the proposed network effectively reduces noise on all four datasets. This can be attributed to the feature enhancement processing and the powerful representation capability to aggregate global and local information. Notably, even with fewer bands and complex pixel distributions, the proposed method can also achieve promising classification maps on the XZ dataset.

3) Monte Carlo simulation: To comprehensively evaluate the performance of different transformer-based methods, we conducted a Monte Carlo simulation experiment. Fig. 5 illustrates the normal distribution curves for the OA of each method, plotted using their respective means and standard deviations.

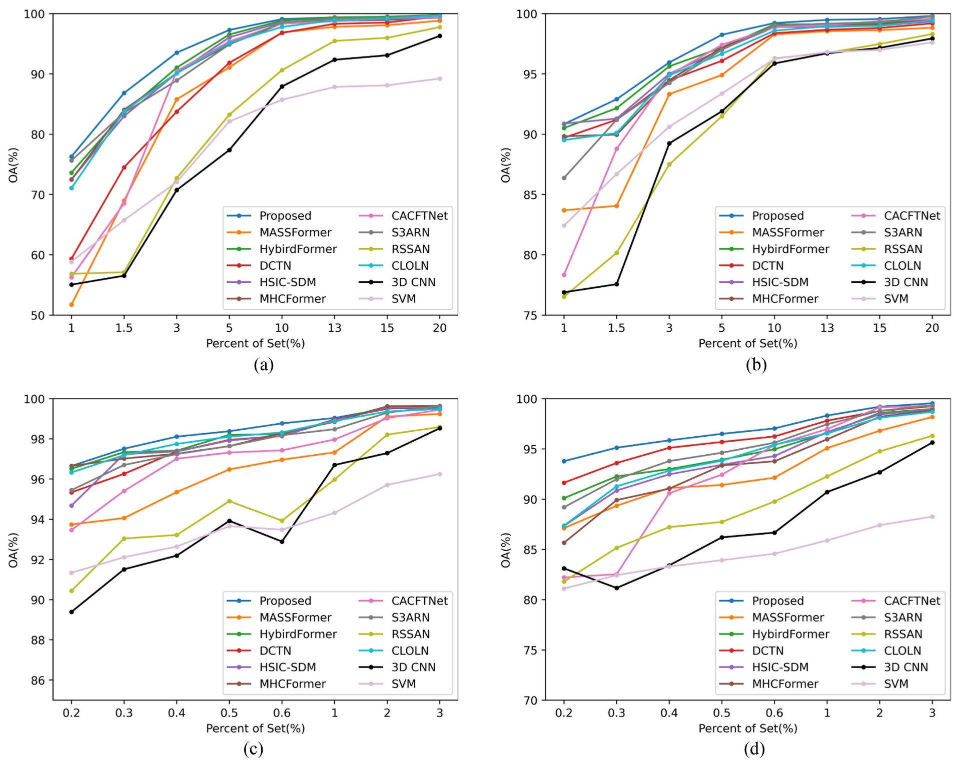

Fig. 11. Classification results of all models under different training set ratios. (a) IP. (b) HU. (c) XZ. (d) HC.

These curves visually provide the performance variability and the central tendency. The x-axis represents the percentage of OA, while the y-axis represents the probability density. It can be seen that HybirdFormer, DCTN, and MHCFormer demonstrate high mean OA values on certain datasets with slightly broader distributions. In addition, both MASSFormer and CACFTNet have lower mean OA values and broader distributions, indicating more variability and generally lower performance compared to the other methods. However, these methods exhibit greater variability in performance, which may impact their reliability in practical applications. Notably, the proposed EATN method shows a relatively high mean OA with a narrow spread, indicating quite stable performance across multiple experiments. This suggests that our EATN method is not only more accurate but also exhibits greater stability under varying conditions.

4) Model Efficiency Analysis: In this section, we evaluated the model based on three matrices: the number of parameters (Params), floating-point operations per second (FLOPs), and inference time (Infer). Here, Infer represents the testing time on the test set, excluding accuracy calculation time and preprocessing time. Table IX shows the efficiency analysis of different transformer-based HSIC methods on the IP datasets. It can be found that DCTN exhibits relatively higher Params and FLOPs. The main reason is that DCTN integrates more 3-D convolutions and self-attention blocks into the classification framework. Although both our EATN method and HybridFormer involve four self-attention operations, our method significantly reduces FLOPs and Params compared to the HybridFormer. This is mainly attributed to the proposed method which performs more efficient and effective self-attention calculations within the IST and SCA modules. MASSFormer performs fewer self-attention operations among all methods, but the proposed method still maintains superior efficiency compared to other transformer-based HSIC methods. Meanwhile, it obtains the best classification performance. Notably, the proposed model exhibits the highest inference efficiency among all methods, making it particularly suitable for real-time classification applications.

E. Hyperparametric Analysis

Due to the different importance of features extracted from the spectral and spatial branches, it is unreasonable to directly combine these features. To address this problem, we introduced the trainable balance operator λ, assigning different weights to these two branches. Fig. 9 shows the loss value of the proposed model with different settings of the balance operator λ on the IP and XZ datasets. The x-axis represents the epoch, while the left and right parts of the y-axis depict the loss value and the correspondingλ value, respectively. In general, a larger λ implies a higher weight assigned to the spectral branch, and vice versa. It can be seen that λ converges to approximately 0.46, indicating that features extracted by the spatial branch are relatively more important for pixel discrimination on the IP dataset. Conversely, the value of the hyperparameterλ approximates 0.55 on the XZ dataset, signifying a larger contribution from the spectral branch. The main reason is that the IP dataset contains more ground object categories and the XZ dataset includes relatively dispersed labeled pixels. By adaptively fusing

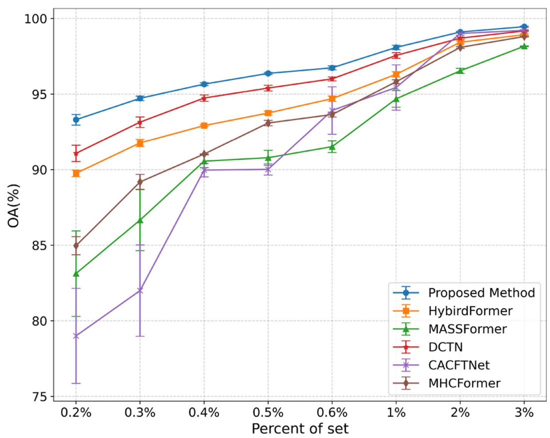

Fig. 12. Mean and standard deviation of the transformer-based models under different training set ratios.

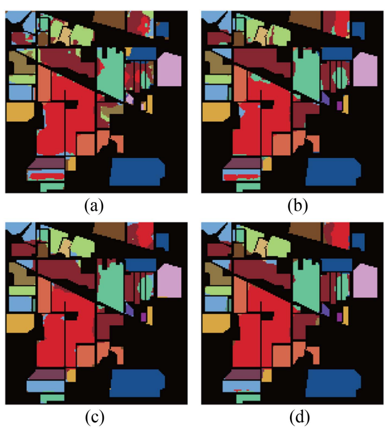

Fig. 13. Classification results under different ablation experiments on the IP dataset. (a) 1. (b) 3. (c) 6. (d) 8.

the features from these branches, the designed network becomes more suitable for various datasets, thereby facilitating practice classification tasks. Notably, the experimental results shown in Fig. 9 also reveal that the loss of both IP and XZ datasets is close to 0 within 50 epochs, further verifying the efficiency of our fusion strategy in promoting the convergence rate of the model.

F. Ablation Study

In this section, a series of ablation experiments were conducted on each dataset to assess the effectiveness of each component in our EATN method. The results are presented in Table X. In Table X, "×" denotes the disabled module, "−" denotes the replaced module, and "✓" indicates its activation.

In the first experiment, we solely used the proposed SSFE module without any feature extraction module. SSFE(N) represents the neighborhood self-similarity descriptor in SSFE, while SSFE(C) represents the center self-similarity descriptor. It can be seen that the performance in this configuration is notably poor, indicating that SSFE fails to independently classify the pixels. To evaluate the effectiveness of our proposed spatial–spectral branch, we designed the second and third experiments, where we introduced the SIT module and the SCA module to the SSFE module, respectively. The purpose is to evaluate the contributions of these feature extraction branches. It can be seen that the performance of the model can be significantly improved by incorporating these branches. In the fourth and fifth experiments, we replaced the GISSA and SWSA modules in SIT and SCA with traditional self-attention modules, respectively. The results indicate that the proposed SIT and SCA modules can reduce the computational burden while positively impacting the overall classification performance. Notably, in the eighth experiment, we fused the features of these branches, further improving classification effectiveness and surpassing that of any individual branch. In addition, we independently evaluated the feature enhancement capability of SSFE in the sixth and seventh experiments. It is clear to see that the classification capability decreases significantly after the SSFE module is removed. In Experiment 7, adding SSFE(N) into the network led to a noticeable improvement in model performance, and further incorporating SSFE(C) resulted in the best classification performance. This phenomenon further confirms the effectiveness of feature enhancement based on original patches.

Fig. 13 exhibits the classification results of Experiments 1, 3, 6, and 8. As shown in Fig. 13(a), Experiment 1 only applies SSFE to the original patches without subsequent spatial and spectral information encoding, resulting in the poorest classification performance. In Experiment 3, the absence of spectral features extracted by SIT leads to poorer edge pixel handling in Fig. 13(b) compared to Fig. 13(d). This may be due to the lack of discriminative spectral features, making the classification more susceptible to noisy pixels. In addition, Fig. 13(c) exhibits noticeable salt-and-pepper noise, which can be attributed to the lack of SSFE enhancement for the original patches, significantly weakening the model's ability to utilize spatial positional cues. These experimental results demonstrate that the proposed model achieves the best classification performance across all four HSI datasets. The effectiveness of each module in our proposed EATN method has been demonstrated in HSIC applications.

G. Influence of Training Set Size

In this section, we evaluated the generalization ability of the proposed method by setting different proportions of training set samples on four datasets. For the IP and HU datasets, we randomly selected 1%, 1.5%, 3%, 5%, 10%, 13%, 15%, and 20% samples as training sets, respectively. In the case of the XZ and HC datasets, we randomly chose 0.2%, 0.3%, 0.4%, 0.5%, 0.6%, 1%, 2%, and 3% samples as training sets, respectively.

Fig. 11 shows the classification results of different methods under different proportion settings of the training set. The best OA from ten independent experiments is reported as the final result. It is clear to see that the classification performance of all methods shows an increasing trend as the proportion of training set samples increases. With sufficient training data, almost deep learning methods achieved near-optimal performance. In contrast, the performance of SVM cannot show a significant improvement with larger training sets, likely due to its limited nonlinear representation capabilities. However, our proposed method consistently outperforms other methods with different proportions. Specifically, when the number of training samples is extremely limited, some HSIC methods encounter a sharp decline in classification performance. However, the proposed EATN method consistently maintains a significant classification performance on four HSI datasets. This demonstrates the powerful generalization ability of the proposed method. Fig. 12 illustrates the means and standard deviations of OA for transformer-based methods on the HC dataset. The vertical lines on each data point represent the standard deviation, indicating the variability in performance between different ratios of the training set. The proposed method consistently exhibits a lower standard deviation and a higher average OA, demonstrating its stability in classification tasks.

V. Conclusion

In this article, a novel EATN is developed for HSIC problems. Considering the high-dimensional property of HSIs, our designed SIT module efficiently reduces the considerable parameter number generated from global self-attention computing. Meanwhile, it facilitates the model to extract more discriminative features through the local–global information interaction. In addition, the SCA module in EATN introduces SWSA between multilayer SCA modules, further alleviating the computational burden of the proposed framework. To enhance feature representation, we employ the SSFE module to process the input patch for subsequent feature extraction. The collaborative integration of these designed modules yields accurate classification results. Experimental evaluations on four widely used datasets demonstrate the effectiveness of our proposed method. Although the self-similarity enhancement strategy has proven effective, it may introduce additional computational costs during training. Consequently, one of our future efforts will be to investigate how to integrate the enhancement strategy into transformer.

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